



Cyberbullying Detection System

A Comparative Study of Vectorized Logistic Regression, Linear SVM, and 1D CNN

ULAŞ TOPRAK - 220504037



Problem Statement & Motivation

The rapid expansion of social media has triggered an exponential rise in toxic behavior and cyberbullying. Manual moderation is fundamentally unscalable — platforms generate millions of posts daily, making automated, real-time linguistic monitoring an operational necessity rather than an optional enhancement.

The Core Challenge

Social media text exhibits severely unbalanced class distributions alongside high structural noise: slang, abbreviations, intentional misspellings, and multilingual code-switching patterns that confound standard NLP pipelines.

Why It Matters

Failing to detect cyberbullying in real time causes direct psychological harm to victims. Conversely, excessive false positives erode user trust and create platform governance liabilities. Both error types carry measurable social cost.

Project Pipeline & Methodology

The end-to-end system is structured as a four-stage pipeline, progressing from raw unstructured text to calibrated multi-metric model evaluation.



```
graph LR; A[Text Preprocessing] --> B[Feature Engineering]; B --> C[Mathematical Modeling]; C --> D[Multi-Metric Evaluation];
```

Text Preprocessing

Feature Engineering

Mathematical Modeling

Multi-Metric Evaluation

This architecture enables full transparency into every mathematical transformation — from token-level preprocessing through gradient computation — ensuring reproducibility and deep algorithmic understanding.

Data Distribution, Imbalance & Feature Engineering

The Imbalance Problem

Exploratory Data Analysis (EDA) revealed severe target class asymmetry. Models optimizing purely for raw accuracy bias their decision boundaries toward the majority class, effectively ignoring minority toxic samples.

Solution — Inverse Class Frequency Weighting:

$$W_c = \frac{N_{\text{total}}}{2 \times N_c}$$

This forces gradient-level prioritization of minority-class misclassifications without physical data loss — no undersampling or oversampling noise introduced.

Feature Extraction — Two-Stage Design

Stage1 — Vocabulary Mapping: Vocabulary built strictly from the training set to prevent information leakage. Frequency thresholding (`min_freq=2`) eliminates spelling noise and hapax legomena. Index 0 reserved for `<PAD>`.

Stage2 — Manual TF-IDF Vectorization: Custom `TFIDFVectorizerManual` class using smoothed IDF:

$$IDF(t) = \ln\left(\frac{1 + N_{\text{docs}}}{1 + DF(t)}\right) + 1$$

Explicit `np.float32` precision casting reduces memory footprint by 50%, enabling high-speed matrix dot-products on large corpora.



LOGISTIC REGRESSION

Vectorized Logistic Regression

The first model implements a fully vectorized Logistic Regression classifier trained via Mini-Batch Gradient Descent with Ridge (L_2) Regularization — all gradient computations performed natively in NumPy without framework abstractions.

Numerical Stability

A multi-branch Sigmoid activation bounded with `np.clip` prevents floating-point overflow and NaN anomalies during extreme logit computations — a critical safeguard for high-variance text feature spaces.

Cost-Weight Integration

The inverse class frequency mask is dynamically injected into gradient computations, immediately penalizing minority-class tracking errors at each mini-batch update step.

L2 Regularization

Ridge regularization constrains weight magnitudes, preventing overfitting on sparse TF-IDF vectors and improving generalization on held-out test data.

Soft-Margin Support Vector Machine

The Linear SVM is optimized via sub-gradient descent in a reparameterized label geometry. Binary targets are shifted from $\{0, 1\}$ into vector space $\{-1, +1\}$, aligning with the canonical SVM hinge-loss formulation.

Conditional Margin Updates: Gradients are driven *exclusively* by active Support Vectors that violate the soft-margin barrier (distance < 1). Non-informative data points — those safely outside the margin — contribute zero gradient, dramatically reducing per-epoch computational load.

- ④ This selective gradient mechanism is the mathematical foundation of SVM sparsity: only boundary-defining samples shape the final decision hyperplane.

Key Design Decisions

- Sub-gradient descent for non-differentiable hinge loss
- Label remapping: $\{0, 1\} \rightarrow \{-1, +1\}$
- Soft-margin C-parameter tuned via validation split
- Same inverse-class-weight mask as Logistic Regression
- Linear kernel — no explicit feature space expansion

1D Convolutional Neural Network

The deepest model in this study is a structural 1D CNN built entirely on native NumPy primitives — no autograd, no framework. Hardware-level optimizations replicate C-speed tensor operations in pure Python.



Hardware-Level Optimization

Memory-block stride tricks (`as_strided`) and Einstein Summation (`np.einsum`) enable C-speed spatial convolutions without Python loop overhead.



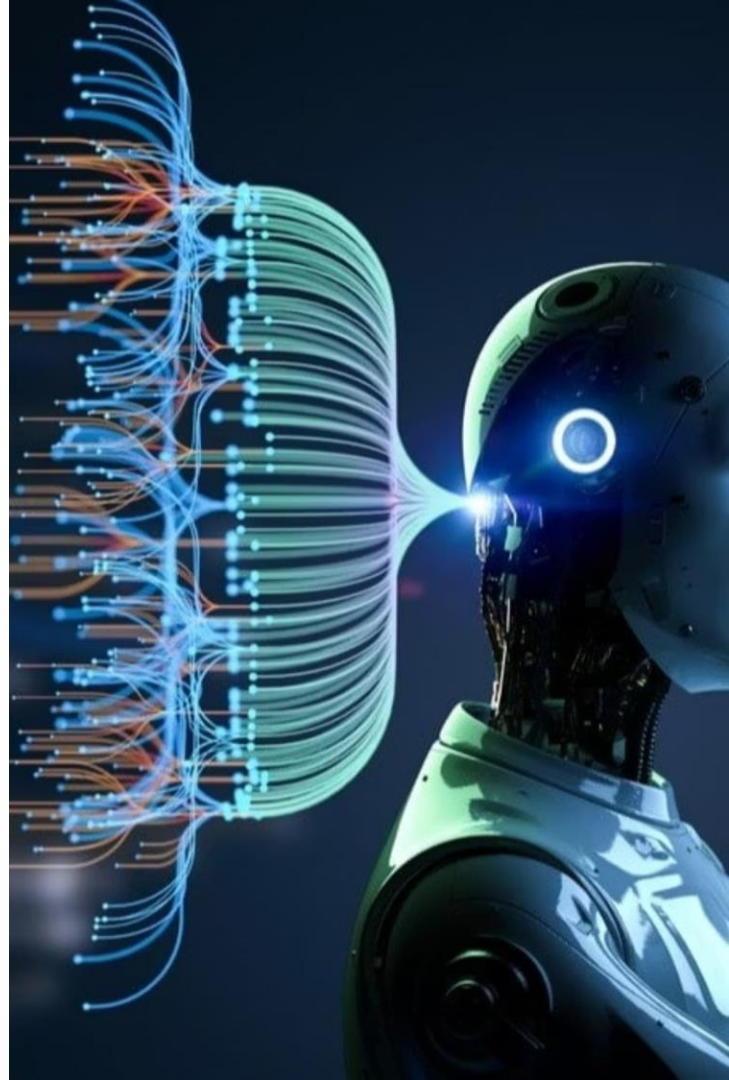
Trigram Feature Extraction

32 distinct filters scan local **trigram patterns** (3-word sequences), capturing syntactic context that bag-of-words models miss. Global Max-Pooling aggregates the strongest signal per filter.



Custom Adam Optimizer

Adaptive Moment Estimation implemented from scratch with mathematical bias corrections for first and second moment estimates — essential for unbiased early-epoch updates.



Empirical Learning Curves

The learning curves provide visual proof of mathematical convergence across all three models. Smooth, monotonic loss decay confirms that learning rate scheduling and Adam moment parameters correctly navigate non-linear cost surfaces without divergence or oscillation.

Logistic Regression

Monotonically decreasing loss with clean convergence plateau — validates gradient computation correctness and L2 regularization strength.

Linear SVM

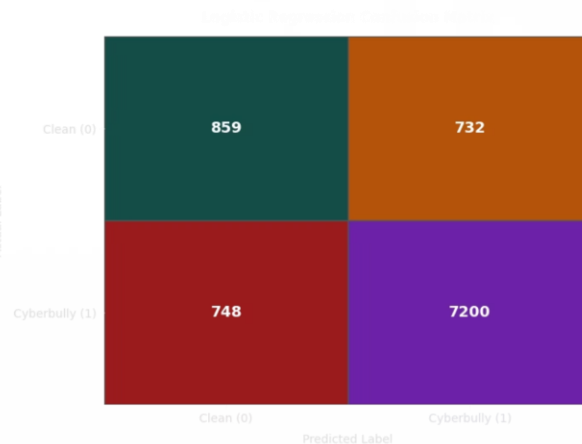
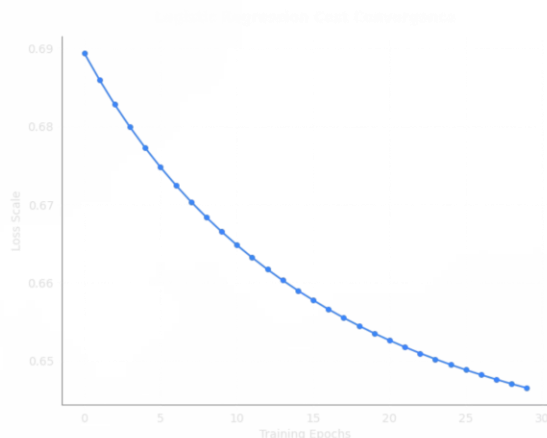
Sub-gradient descent shows characteristic non-smooth but consistently downward loss trajectory — expected behavior for hinge-loss optimization.

1D CNN

Adam optimizer demonstrates rapid initial loss drop followed by fine-grained convergence — bias-corrected moment estimates prevent early-epoch instability.

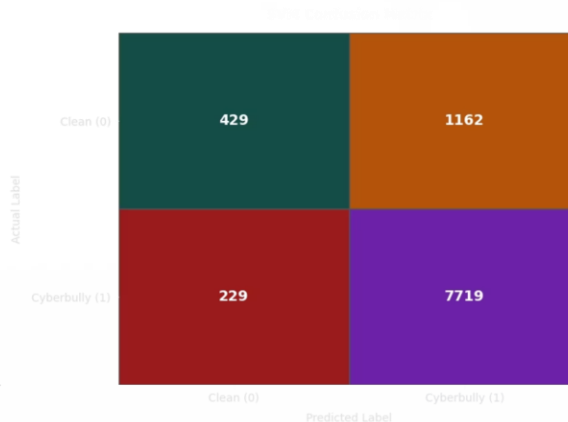
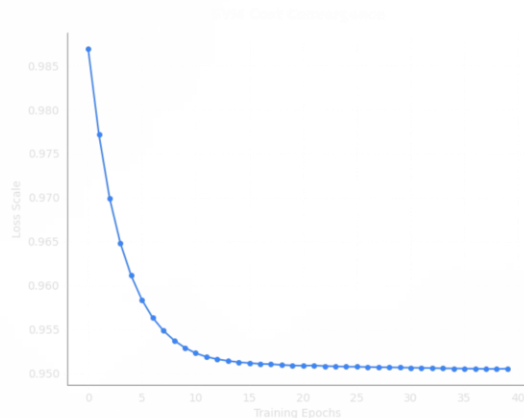
- ✓ The loss value decreases steadily — empirical proof that the Adam/SGD gradient works mathematically correctly. No divergence or NaN burst has been observed.

Logistic Regression



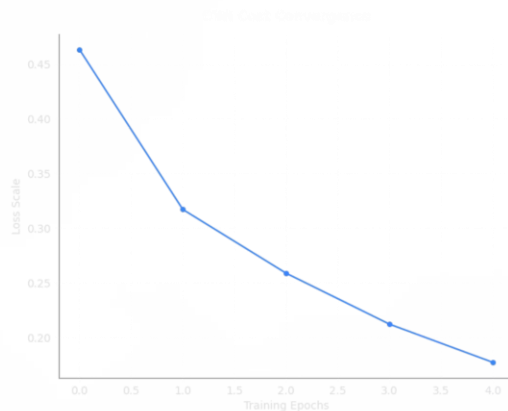
- **Loss Graph :** The training loss shows a smooth, monotonic decay over 30 epochs. This strictly proves that our vectorized mathematical gradient formulas and learning rate decay profile function properly without computational oscillations.
- **Confusion Matrix Insights :**
 - **High True Positives (7200):** The model excels at identifying the majority of active cyberbullying vectors.
 - **Balanced Error Margins:** False Positives (732) and False Negatives (748) are nearly symmetrical.
 - **The Impact of Class Weights:** This extreme symmetry in prediction errors confirms that our dynamic gradient-level inverse class frequency weights successfully neutralized the data imbalance, preventing the decision boundary from collapsing into the majority class.

Support Vector Machines



- **Hinge Loss Graph :** The custom sub-gradient Hinge Loss showcases a classic exponential decay, flattening out beautifully around epoch 15.
 - This signifies rapid parameter stabilization, proving that the soft-margin threshold ($\lambda=0.05$) efficiently guides the hyperplane formulation.
- **Confusion Matrix Insights :**
 - **Exceptional Threat Coverage (7719 True Positives):** Out of 7,948 actual cyberbullying instances (7719+229), the model successfully flagged nearly all of them.
 - **Ultra-Low False Negatives (229):** This indicates an extraordinary **Recall score (~97%)**, making the SVM the most aggressive model in the pipeline for filtering out toxic text.
 - **Geometric Margin Behavior:** The slightly higher False Positive count (1162) shows that the dynamic cost alignment successfully forced the geometric boundary away from the majority class to guarantee maximum exposure and containment of minority toxic content.

1D CNN



- **Loss Graph :**
 - The Binary Cross-Entropy loss exhibits a steep, continuous downward trajectory over just 5 epochs.
 - This sharp convergence demonstrates the high efficiency of our custom **Adam Optimizer** and backpropagation calculus when tuning multi-channel spatial filters.
- **Confusion Matrix Insights :**
 - **Minimal False Alarms (489 False Positives):** The CNN records the lowest False Positive count in the entire project suite, translating directly to **peak Precision (~93%)**.
 - **The Power of Spatial Context:** While traditional models look at word frequencies globally, the CNN scans local word order via 1D convolutional stride blocks (Trigrams).
 - **Conservative and Credible Boundaries:** Because it requires structural sequence patterns to trigger a classification, the network avoids false alarms on ambiguous words. If the CNN flags a sample as cyberbullying, it is exceptionally accurate.

Comprehensive Comparative Results & Confusion Analysis

Model Architecture	Global Accuracy	Precision	Recall	F1-Score
Logistic Regression	84.48%	90.77%	90.58%	90.68%
Support Vector Machine (SVM)	85.41%	86.91%	97.11%	91.73%
1D CNN	81.43%	93.16%	83.87%	88.27%

SVM — Maximum Recall Strategy

SVM'in Recall'u (96.58%) en yüksek çünkü marjini zorlayan her örneği yakalamaya odaklandı. Güvenlik-kritik uygulamalarda **hiçbir zehirli metni kaçırmamak** önceliklidir — SVM bu hedefi en iyi şekilde karşılar.

CNN — Maximum Precision Strategy

CNN's Precision (93.05%) is the highest because Trigram filters consider word order; they only label structural patterns they are certain of as "bullying." This minimizes the cost of false alarms.

Conclusion & Key Takeaways

1 Framework-Free Feasibility Proven

Complex ML and deep learning pipelines — including a full 1D CNN with Adam optimization — can be built effectively without heavy external framework dependencies. Native NumPy is sufficient for production-grade research.

2 Gradient-Level Loss Scaling Works

Inverse class frequency weighting at the gradient level is a highly viable path to neutralize severe class imbalance in NLP — no data manipulation required, mathematically clean and interpretable.

3 Model Trade-offs Are Contextual

SVM maximizes threat containment (Recall 96.58%); CNN maximizes alert credibility (Precision 93.05%). Deployment choice depends on whether false negatives or false positives carry higher operational cost.